

NIRNAY: A 450M Calibrated Decision Model with Stable Product-Quantized Concepts

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October 5, 2026

Abstract

Small decision models turn a state plus a typed question into calibrated probabilities in one forward pass. They are cheap to serve but hard to train: we document two distinct training-collapse mechanisms in a product-quantized concept bottleneck, both ending in constant predictions (1/77 accuracy on Banking77). First, the encoder output grows unboundedly (RMS 0.089 healthy, 13.5 dead) while codebook centroids stay near zero, so the vector-quantization loss explodes 300 $\times$. A directional test shows descending the task loss raises the scale while descending the quantization loss lowers it. Second, code usage collapses to one code per chunk (dominant share 0.97) with no loss spike at all, and the commitment term then homogenizes encoder features. The fixes are small and structural: an affine-free LayerNorm on the quantized input, which removes scale as a degree of freedom, and a load-balancing auxiliary loss over hard assignment fractions times soft assignments, whose gradients cannot saturate at collapse. A failed entropy-based alternative is reported. A heldout-probe safety net (periodic eval, best-checkpoint retention, collapse abort) caught four real collapses during development. The resulting 450M model scores **0.8792** on Banking77 test (3,080 cases, Brier 0.208, fitted ECE 0.045), against 0.803 for the Jev 1.13.0 API on the same cases (ours fine-tuned on the train split; Jev zero-shot) and 0.64 for the 144M Julia-1 on its easier 72-label pilot. Code, weights, and raw eval JSONs are public under Apache-2.0 at <https://github.com/eulogik/nirnay> and <https://huggingface.co/eulogik/nirnay-450m>.

Keywords: decision models, calibration, vector quantization, concept bottleneck, load balancing, Banking77, small language models.

1 Introduction

Production systems need answers, not chat: which team owns this incident, is this charge a duplicate, how urgent is this request. Decision models take a state plus a typed question (choice, score, yes/no) and return probabilities in one pass. The category has a clear leaderboard (JevBench [14]) and fast-moving open entries (Laya, Julia-1, CLM-8B), but small open models keep failing in the same places: many options, honest calibration, and stable training.

This paper reports what broke when we trained one, and the fixes that held through a full 7,000-step run. Our contributions:

- Two collapse mechanisms with opposite signatures, isolated by feature-path probes: unbounded encoder-output scale (loss spike 300 $\times$) and code-usage collapse (no spike, constant predictions either way).

- A directional-gradient test showing the task loss climbs scale while the quantization loss pulls it down, which explains the runaway.
- Two structural fixes: affine-free LayerNorm before quantization, and a saturation-proof load-balancing term. Plus a documented failed attempt (entropy deficit) with the reason it fails.
- A probe safety net that bounds collapse damage to minutes and keeps the best checkpoint automatically.
- A 450M Apache-2.0 model at 0.8792 Banking77 test accuracy with fitted ECE 0.045, with every number reproducible from shipped artifacts.

2 Background

Vector quantization maps continuous states to a finite codebook [1]. Concept bottleneck models route predictions through human-readable concepts [2]. Load-balancing losses of the form $K \sum fP$ keep mixture assignments spread [3]. Scalar quantization (FSQ [9]) avoids codebooks entirely; variational bottlenecks constrain representations explicitly. For small models, process-level supervision beats outcome-only rewards [11], and outcome-only rewards fail on long-context grounding [12]. Activation distributions decide quantizer behavior [10], and VQ collapse is a property of the spanned space rather than single vectors [8]. Discrete concepts can also be read off frozen states post hoc [7].

We train with AdamW [4]. Calibration is reported as raw and fitted expected calibration error on every eval, never one alone. Our reinforcement step follows the calibrated-decision recipe (RLCD) [19]. The eval landscape: JevBench v1.5.4 ranks 106 systems over 1,624 decisions (904 open, 720 sealed) on intelligence, calibration, speed, and cost; on 2026-09-30 the board reads Cygnet 73.7, Winnow-12B 73.2 (intelligence 74.4, sealed 76.4), Jev 1.13.0 72.1 (intelligence 72.0, calibration 88.0). Banking77 [5] is 77-way intent classification (3,080 test cases). Our base is Laya, a 421M Apache-2.0 decision encoder.

3 Model

A frozen ModernBERT-style encoder [6] (LoRA adapters trainable) maps token and byte inputs to hidden states h . A concept bottleneck encodes $z = \text{encode}(h)$ (128 dims), quantizes four 32-dim chunks against four 32-code codebooks, mixes $M=4$ slots, and adds a zero-initialized residual δ back to h . A 2-layer head scores dense markers; a coarse-to-fine pointer re-ranks the top 20 for 77-way rows. Per-bucket temperatures calibrate the output. Deep supervision at layers 4/8/12 and the RLCD phase exist only in training.

The plan loss is fixed:

$$L = L_{\text{choiceCE}} + L_{\text{RPS}} + L_{\text{BCE}} + L_{\text{rel}} + 0.3 L_{\text{NCP}} + 0.2 L_{\text{deep}} + \lambda L_{\text{calCE}}, \quad \lambda = 0.005, \quad (1)$$

with $L_{\text{NCP}} = \text{codebook} + \text{commitment} + \text{gateUsage}$. The optimizer has three groups: lr 5×10^{-4} (rest), 10^{-5} (pretrained head/scorer/embeddings, 26.2M params), and 10^{-4} (concepts, 4.7M). Global gradient clip 1.0. Mix: 8,412 train / 2,103 heldout (Banking77 train split plus 512 synthetic), batch 4, LoRA rank 8, seed 13, one M4 Mac.

4 Collapse diagnosis

Two supervised runs died at exactly 1/77 test accuracy with different loss signatures. A feature-path decomposition (stratified $n=308$) exonerated head, scorer, and delta (each scores ≈ 0.53 on healthy encoder features) and convicted the encoder adapters (feature std 0.44 healthy, 0.18 dead; zeroing LoRA restores features).

Mechanism A: scale runaway. Table 1 tracks the encoder-output RMS on fixed probe inputs. Centroid RMS stays near 0.04 throughout, so $\text{mse}(z, z_q)$ tracks z and the NCP term explodes (22.3 at step 150 in one run, 323.9 at step 75 in another, 31.5 at step 6416 in the 7,000-step run). Hidden-state RMS stays near 1.13 and encoder weight RMS stays at init, so the growth is a gain/regime shift inside the projection (GELU mean 0.035 to 0.529), not input or weight blowup. A directional test at a pre-death checkpoint settles the cause: a unit step along the task gradient *raises* z std (+455 per unit), while a unit step along the NCP gradient *lowers* it (−2980). Scale is task-driven; the codebook chases it at full learning rate; the loop has no fixed point. Figure 1(a) shows the progression.

Table 1: Encoder-output RMS on fixed probe inputs. Centroids stay near 0.04; hidden RMS stays near 1.13.

Checkpoint	z RMS	centroid RMS	hidden RMS	NCP
Healthy (250 steps)	0.089	0.042	1.141	0.03
Pre-death (25 steps)	0.958	0.039	1.133	1.70
Dead (50 steps)	5.085	0.036	1.148	41.98
Dead (75 steps)	13.494	0.043	1.130	323.89

Mechanism B: usage collapse. With no pressure on code usage, the per-chunk dominant share climbs (healthy 0.27 to 0.49 across 32/32 codes; dead 0.92 to 0.99 on 2 to 14 codes). The quantized states go near-constant and commitment homogenizes encoder directions. Crucially, a run with the scale fix but no usage term still dies at step 75 with NCP flat near 1.9: usage collapse alone is sufficient, and it gives no loss warning. Figure 1(b) shows the progression.

5 Fixes

LayerNorm on z before quantize (affine-free, zero params, zero state-dict keys, so old checkpoints still load). Scale leaves the system: the task gradient can only rotate z , NCP stays $O(1)$, and assignments become direction-based. This directly removes mechanism A.

Load-balancing aux term. Let f be hard assignment fractions (detached) and P the mean soft assignment at temperature $\tau = 0.1$:

$$L_{\text{usage}} = w \sum_{\text{chunks}} \max(K f \cdot P - 1, 0), \quad w = 0.01. \quad (2)$$

It is linear in P , so gradients flow even at collapse. The first attempt, an entropy deficit $w(\ln K - H)$, failed A/B testing twice: at $\tau = 0.5$ it is too soft to see early skew (gaps near 0.08), and at sharp τ it saturates to one-hot exactly when pressure is needed most. The failed form is reported so nobody repeats it.

Safety net. Training splits into segments with a heldout probe after each (same argmax rule as final eval), recording accuracy plus per-chunk unique/mode statistics. Best checkpoint is retained;

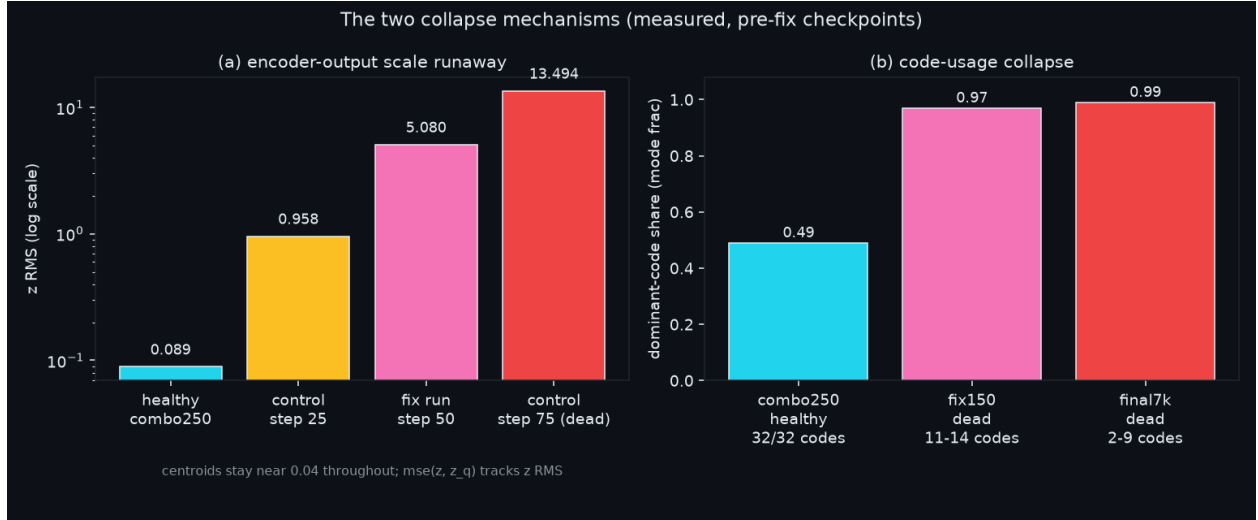


Figure 1: Measured pre-fix signatures. (a) Encoder-output scale runaway on a log axis. (b) Code-usage collapse as dominant-code share.

training aborts when probe accuracy halves or NCP spikes. This caught four real collapses during development and bounds any future death to minutes.

6 Experiments

A/B ladder on the known-kill config (concepts-lr 5×10^{-4}), same seed throughout (Table 2). The full fix is the only variant that both keeps usage healthy and learns; the shipped config (concepts-lr 10^{-4}) then reproduces the pre-fix baseline exactly (test 0.5218 vs 0.5195).

Table 2: Fast-repro A/B, kill config unless noted. Acc = heldout probe.

Variant	Outcome	Best acc	Usage at end
No fix (control)	dead at 75 (NCP 323)	0.25	collapsed
LN only, no aux	dead at 75, NCP flat	0.22	collapsed
LN + entropy aux	dead at 50	0.32	collapsed
LN + balance aux	abort at 100, peak 0.40	0.40	healthy
LN + balance aux, shipped lr	250/250 clean	0.51	healthy

Full run. 7,000 steps with probes: accuracy 0.28 to 0.89 (best 0.8906 at step 5000, final 0.8789), NCP flat between 1.91 and 2.01 for all 7,000 steps, usage oscillating in a stable band with the aux term pushing back every narrowing (Figure 2). The run passed straight through the step-6400 zone that killed its predecessor. Loss 3.93 to 1.25.

Banking77 test (3,080 cases, Table 3 and Figure 3). Phase A: 0.8656 accuracy, Brier 0.240, raw ECE 0.110, fitted 0.033. Plus 50 RLCD steps (Phase B): **0.8792**, Brier 0.208, raw ECE 0.089, fitted 0.045. The best-probe checkpoint scores lower on test (0.8542) than the final one (0.8656): probes detect collapse, they do not fine-rank. The Jev reference ran the same 3,080 cases [15]. On macro-F1, the metric of Cloudflare’s Decision Index 0.2.1 suite, phase_b scores 0.8785 on the same 3,080 cases (per-class scores ship with the release); their self-reported row reads Clef 0.942, Clef-

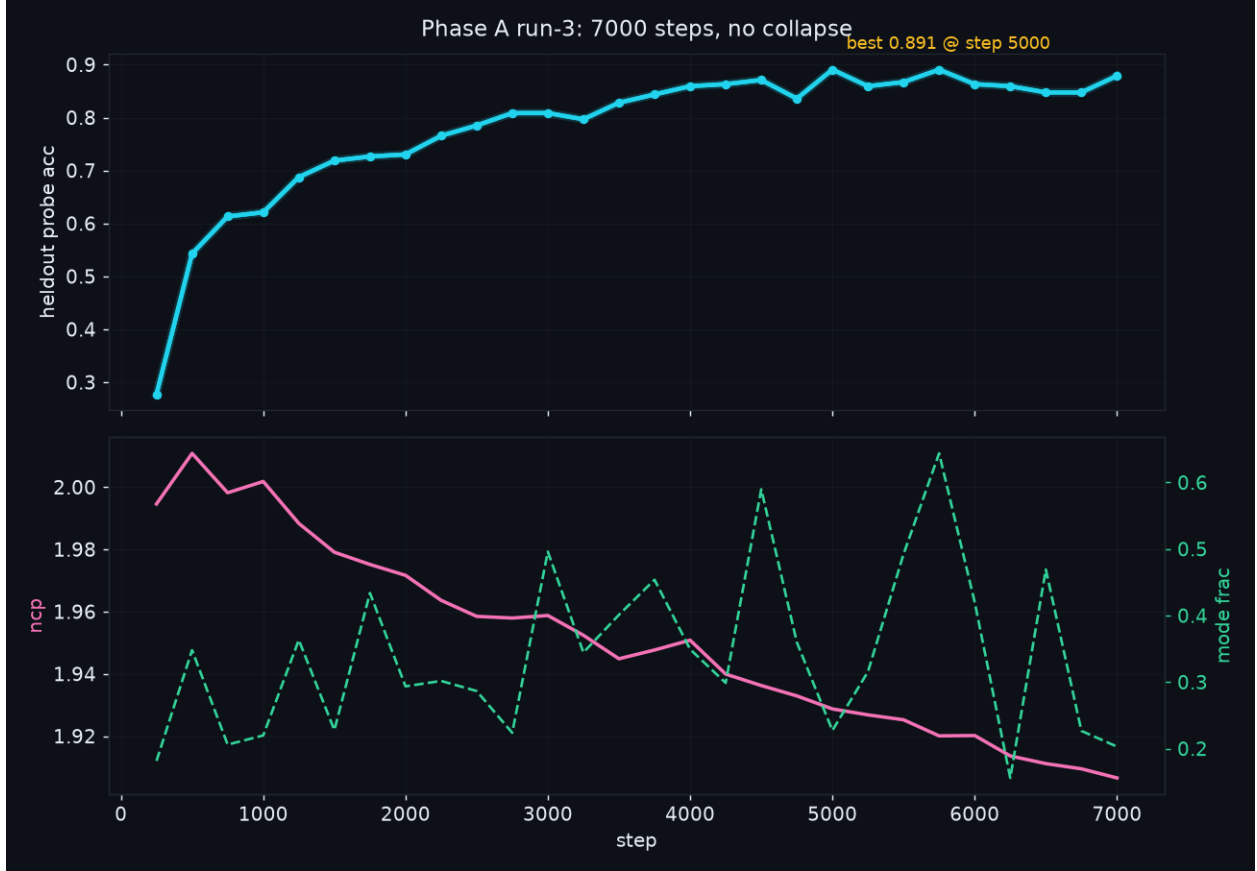


Figure 2: Protected 7,000-step run: climbing probe accuracy, flat NCP, usage in a self-limiting band. No storm at the historical death zone.

Flash 0.909, Kev 0.848, Jev 0.797. Same metric family, different harnesses, so this is a cross-check, not a leaderboard: we beat Jev by 8 points and Kev by 3 at 60x and 20x fewer parameters.

Table 3: Banking77 test (n=3080). Setups stated exactly.

Model	Accuracy	Setup
NIRNAY phase_b	0.8792	fine-tuned 450M, this work
NIRNAY phase_a	0.8656	fine-tuned 450M, this work
Jev 1.13.0	0.803	zero-shot API, same 3,080 cases
Julia-1 144M	0.64	their 72-label pilot (n=100, shortlist)
Untrained	0.1429	fresh weights, our template

Calibration (Figure 4). Most mass sits above 0.9 confidence at 92% accuracy there; fitted ECE 0.045. Raw and fitted are always reported together.

Honest misses. JevBench public (231 items, shipped checkpoint): 0.550 overall (easy 0.875, original 0.569, hard 0.396). Hard states average 1,037 tokens and 63% exceed the 320-token state budget; fitting items still score only 0.488, so context and reasoning both gap. A head-plus-tail windowing fix scored 0.300 against 0.343 head-only and was rejected. A 10k-item programmatic reasoning set (ground truth by construction, nothing copied from JevBench) is built and validated

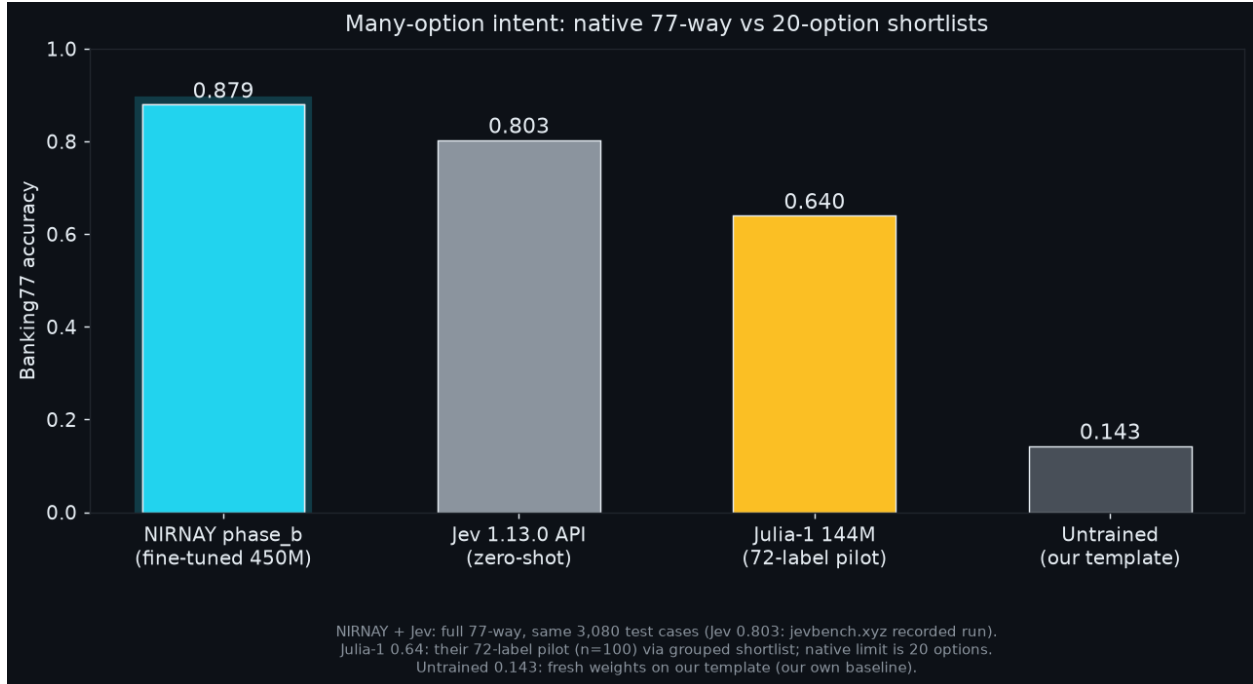


Figure 3: Many-option intent: native 77-way against shortlists. Footnotes in figure state every setup difference.

but untrained at scale.

7 Related work

VQ for concepts appears as post-hoc discovery on frozen states [7] (with LayerNorm around the quantizer, matching our fix), as scalar levels without codebooks [9], and as variational bottlenecks over human concepts [2]. Collapse is analyzed as a property of the spanned space [8]; activation shaping is studied as distribution design [10]. For small-model RL, process rewards beat outcome-only rewards at 494M parameters [11], and outcome-only rewards fail long-context grounding [12]; our Phase B reward is outcome-only, which bounds what it can teach. Positioning, all numbers third-party and dated 2026-09-30: JevBench v1.5.4 ranks Cygnet 73.7, Winnow-12B 73.2, Jev 72.1 over 106 systems [14]; Laya (421M, Apache-2.0, our base [16]) with 33ms inference; Julia-1 (144M, Apache-2.0 [17]) beats Jev on typed decisions but reports 0.64 on its Banking77 pilot and caps native calls at 20 options; CLM-8B (Stanford/NVIDIA, Apache-2.0 [18]) plays a GPU tool-calling lane. Our lane is many-option CPU classification with honest calibration, where the board is thinnest.

8 Limits

512-token context; 361ms CPU per decision in PyTorch (no ONNX yet, no GGUF yet); JevBench-hard reasoning as reported; best-probe checkpoints do not necessarily score best on test.

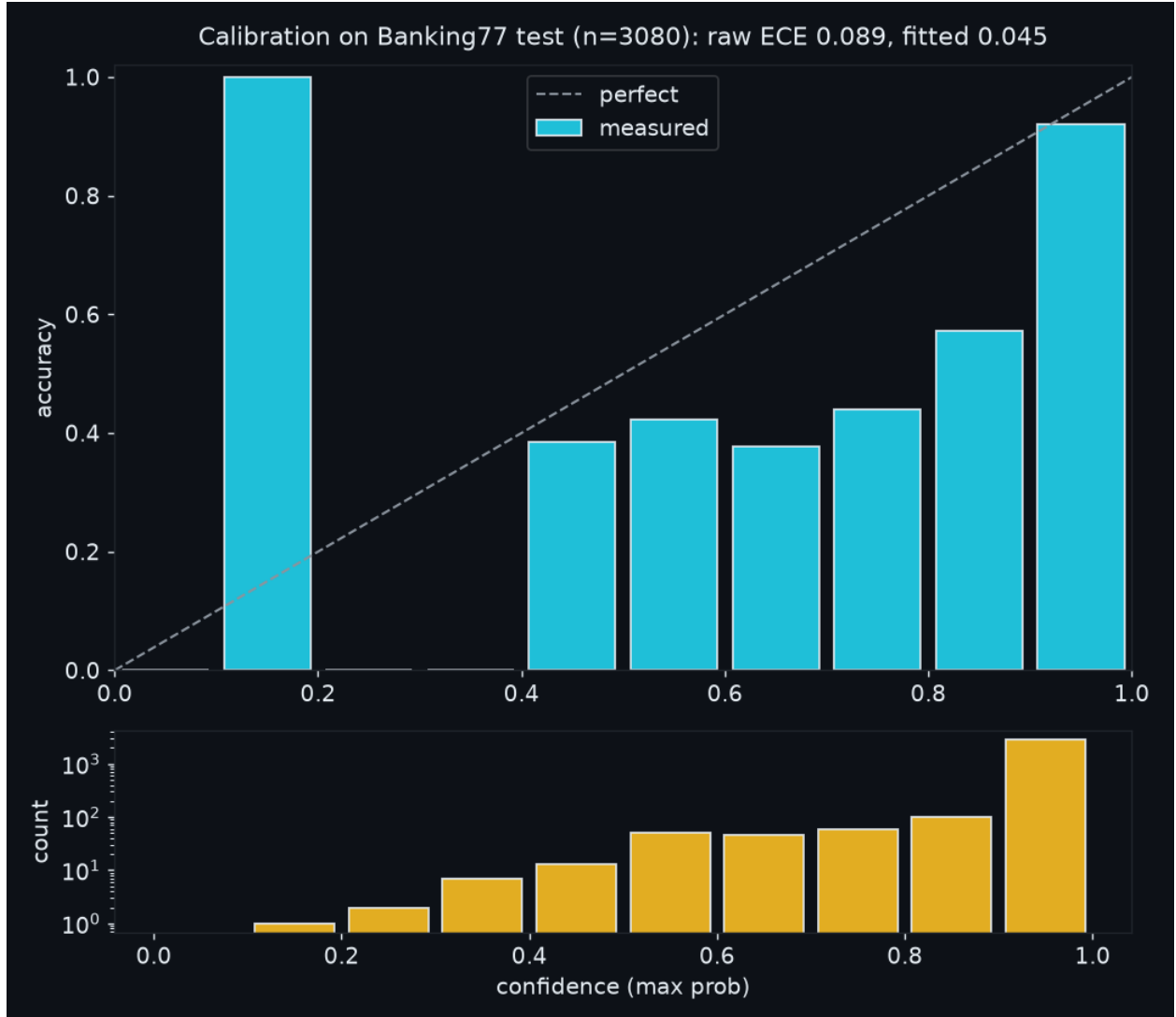


Figure 4: Reliability on Banking77 test (n=3080), measured forward pass. Raw ECE 0.089, fitted 0.045.

9 Reproducibility

Code (<https://github.com/eulogik/nirnay>), weights (<https://huggingface.co/eulogik/nirnay-450m>, phase_b, Apache-2.0), frozen data hashes, all gate checks (19/19), and raw eval JSONs ship with the release. Seeds fixed (13); hardware one M4 Mac. Nothing in this paper comes from closed APIs. Delivered-error claims rest on external ground-truth audits only [13].

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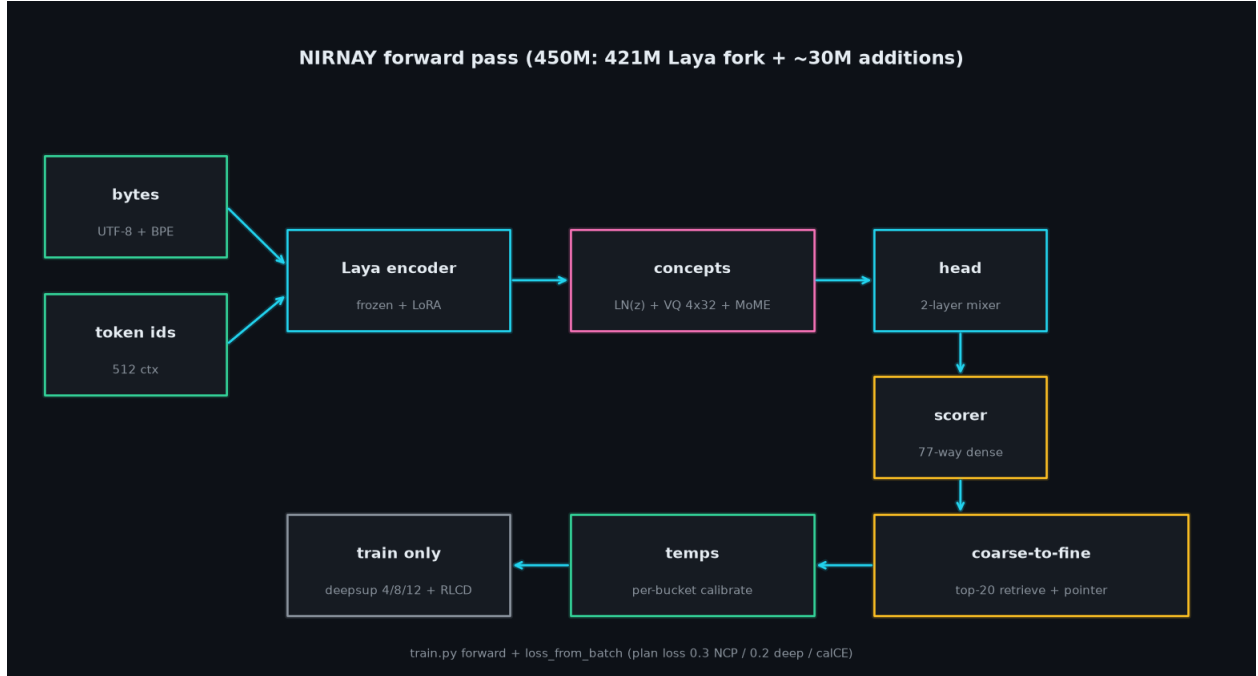


Figure 5: Forward pass: byte path, frozen encoder plus LoRA, concept bottleneck, dense scorer, coarse-to-fine pointer, per-bucket temperatures. Deep supervision and RLCD exist only in training.

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